

Lec 0  
Lec 72

algo & gate

Intro DP  
↳ Lec 71  
Lec 72

Lec\_73

GATE & Problems of DP

In GATE :

60-65%

(done)

WAPs

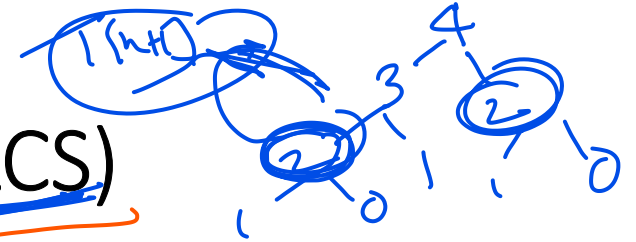
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Competitive coding

LeetCode, C++, C++  
HackerRank

- Fibonacci Series
- Longest common sub sequences (LCS)
- 0/1 knapsack problem
- Sum of subsets
- Matrix chain Multiplication
- Floyd War shall Algorithm

1. fibonacci series ✓



## 2. Longest common sub sequences (LCS)

A subsequence is a sequence that can be derived from the given string by deleting some or no elements without changing the order of the remaining elements.

For example, 'abe' and bd are the subsequences of "abcde"

$S = abcde$   
 $S_1 = abe$   
 $S_2 = bd$

Q1: Which are the subsequences of  $S = \{ABBABB\}$

$s_1 = BBBB$  ✓

$s_2 = BABA$  ✗

$s_3 = AA$  ✓

$s_4 = ABAB$  ✓

$s_5 = BAAB$  ✗

$S = ABBABB$   
X X X X X

$S = ABBABB$   
X X X X X

$S = ABBABB$   
X X X X X

$S_1 = BBBB$   
 $S_2 = BABA$  ✗

What is the meaning of LCS: Longest Common Subsequence

- Our task: Given two strings, s1 and s2, the task is to find the length of the Longest Common Subsequence

Ex1: Input: s1 = "ABC", s2 = "ACD"

Output: ? AC

Length = 2

Explanation:

$S_1 = \{ \textcircled{A}, B, \textcircled{C}, AB, \textcircled{AC}, BC, ABC \}$   
 $S_2 = \{ \textcircled{A}, \textcircled{C}, D, \textcircled{AC}, CD, AD, \textcircled{ACD} \}$

$S_1 = ABC$   
 $S_2 = ACD$

Ex2: Input:  $s_1 = \text{"AGGTAB"} \text{, } s_2 = \text{"GXTXAYB"}$   
Output: ?

$s_1 = A \underline{G} \underline{G} \underline{T} \underline{A} \underline{B}$   
 $s_2 = \underline{G} \underline{X} \underline{T} \underline{X} \underline{A} \underline{Y} \underline{B}$

length of LCS:  
5-6 10

Ans = GTAB 4

Ex3:  $x = \text{bababa}$ ,  $y = \text{abbabaa}$  how many LCS are there and also what is the length

$n=2$

$x = \text{bababa}$   
 $y = \text{abbabaa}$

$s_1 = \text{ababa}$   
 $s_2 = \text{bbab}$   
 $s_3 = \text{baaba}$   
 $s_4 = \text{babba}$   
 $s_5 = \text{bubaa}$   
 $s_6 = \text{babab}$

$s_1 = \text{babaa}$   
 $s_2 = \text{babba}$   
 $s_3 = \text{bbaba}$

$(5)$   
 $(5)$

$x = 10$   
 $y = 15$

$\text{LCS} = 3$

$\text{LCS} = 5$

$\text{LCS} = 7$

$F=6$

Ex4:

# GATE | GATE-CS-2014-(Set-2) | Question 47

Consider two strings  $A = "qpqrr"$  and  $B = "pqprrrp"$ . Let  $x$  be the length of the longest common subsequence (not necessarily contiguous) between  $A$  and  $B$  and let  $y$  be the number of such longest common subsequences between  $A$  and  $B$ . Then  $x + 10y = \underline{\hspace{2cm}}$ .

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$n = \text{length of LCS}$

$y = \text{no. of LCS} \mid n + 10y = ?$

$s_3 = \underline{qp}$

$\text{length}(n) = 4$

$A = \underline{qpqr}r$  (5)  
 $B = p\underline{qprr}p$  (7)

$s_1 = q, s_2 = qp, s_3 = \underline{qp}$

$s_4 = \underline{qpqr} \Rightarrow \text{LCS}$

$y = 3$

$4 + 10(3)$   
 $\Rightarrow 4 + 30$   
 $= 34$

$A = \underline{qpqr}$

$A_1 = \underline{pqr}$

$A_2 = \underline{qrr}$

$A_3 = \underline{qprr}$

$A_4 = \underline{qpqr}$





How to find LCS?

Using algorithm / Coding







